

JOINT-SPARSE OLS MICROPHONE ARRAY DESIGN FOR FREQUENCY-INVARIANT BEAMFORMING

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ABSTRACT

Frequency-invariant (FI) beamforming is essential in broadband microphone array processing to enhance target signals while meeting predefined spatial constraints. To alleviate hardware complexity and processing bottlenecks in dense arrays, designing thinned, sparse array topologies has become a crucial area of research. While our prior work utilized the classical orthogonal matching pursuit (OMP) technique for sensor selection, this paper introduces a joint-sparse extension of the orthogonal least squares (OLS) algorithm as a powerful alternative. We tailor the OLS framework to accommodate joint sparsity constraints across multiple frequency bins and develop an efficient recursive implementation that significantly reduces computational overhead. Numerical simulations on concentric circular arrays (CCAs) demonstrate that while both OLS and OMP provide comparable performance in terms of FI-beampattern fidelity and array gain, the proposed OLS framework offers superior flexibility for geometric layout optimization in highly constrained array geometries.

Index Terms— Orthogonal least-squares, frequency-invariant beamformer, sparse array.

1. INTRODUCTION

The objective of frequency invariant (FI) beampattern (BP) design is to select the weights for sensors across all frequencies in order to minimize the difference between the designed BP and the desired BP, while adhering to certain design constraints, such as the white noise gain (WNG) and a specified suppression level [1]. This goal necessitates an array of closely spaced sensors to effectively meet the BP requirements at high frequencies, while also maintaining a sufficiently large aperture to accommodate the BP needs at low frequencies. These conflicting requirements often result in a large array with many densely packed sensors, which in turn demands increased hardware, processing, and computational capabilities. An alternative to using all sensors in the array to meet the FI-BP requirements is to explore a sparse array design.

In sparse design, we select a small number of sensors that satisfy the design constraints, resulting in an approximate desired BP [2]. Two possible approaches to choose the sensors [3] are: (1) solving the exact sparse optimization problem in a greedy approach using pursuit algorithms (e.g., orthogonal matching pursuit (OMP)), or (2) providing an approximate solution by considering a relaxed sparse optimization. Since the BP needs to be the same across all frequencies, it becomes a joint sparse design problem in which the selected

sensors must satisfy the constraints at all frequencies. Other approaches include genetic algorithms [4, 5], modified particle swarm optimization [6], and biogeography-based optimization [7], which focus on the narrowband case, while broadband sparse designs are presented in [8–12].

Herein, we extend our published works on greedy joint sparse design based on the OMP method [13, 14] and modify the standard OLS to solve the joint sparse optimization problem of designing an FI-BP that is close enough to a predefined desired FI-BP, and still satisfying the WNG and given suppression levels. We modify the selection step in the original OLS for the current joint-sparsity problem by jointly selecting the same set of sensors. Further, to ensure that closely spaced sensors are not selected, we use a mask matrix around each selected sensor so that close sensors are not chosen in future iterations, as used in [13, 14]. At the end of each iteration, we select the sensor that yields the largest decrease in the fit error between the desired FI-BP and the greedy-designed BP while satisfying all constraints. We also offer a recursive implementation, which significantly reduces the computational cost of the OLS method.

To demonstrate the greedy method, we consider concentric circular array (CCA) in our simulations, which are used in several speech applications [15–17]. Simulation results show that the proposed design yields a frequency-invariant (FI) beamformer with high white noise gain (WNG) and directivity. In addition, it requires reasonable computational resources, enabling a practical design for applications involving large arrays with hundreds of candidate sensors.

2. PROBLEM FORMULATION

Consider a planar grid with M possible positions for locating sensors in the array. Let θ denote the azimuthal direction of arrival (DOA) of signals towards the array, measured anti-clockwise from the x axis. The azimuthal DOA of the desired source signal to the array is denoted by θ_{ds} . Let $\mathcal{B}_d^{\theta_{ds}}(\theta)$, $\forall \omega_j \in \Omega$ denote the desired far-field FI BP in the bandwidth of interest Ω , with a mainlobe steered to a specified azimuth θ_{ds} .

In practice, the mainlobe and sidelobe regions may require different resolutions or accuracies. Thus, we set $P' \leq P$ directions that cover the mainlobe region Θ_m , where the subscript m stands for mainlobe, and the remaining $P - P'$ directions that cover the sidelobe region Θ_s , where the subscript s stands for sidelobe. As a particular case, one can set $P' = P$ and treat the entire azimuthal space uniformly. Define the $P' \times 1$ vector $\mathbf{b}_d^m(\theta_{ds}) = [\mathcal{B}_d^{\theta_{ds}}(\theta_1), \dots, \mathcal{B}_d^{\theta_{ds}}(\theta_{P'})]^T$ to be a vector containing samples of the desired BP in directions that cover the mainlobe. Similarly, define the vector $\mathbf{b}_d^s(\theta_{ds})$ associated with the sidelobes.

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The desired FI BP is the $P \times 1$ vector

$$\mathbf{b}_d(\theta_{ds}) = \begin{bmatrix} \mathbf{b}_d^m(\theta_{ds}) \\ \mathbf{b}_d^s(\theta_{ds}) \end{bmatrix}. \quad (1)$$

To obtain such an FI BP, we need to assign frequency-dependent weights to the sensors in the array. Let $\mathbf{h}(\omega_j)$ be an $M \times 1$ weight vector for a given frequency $\omega_j \in \Omega$, $j = 1, \dots, J$, where Ω is the frequency band, and the m th element of $\mathbf{h}(\omega_j)$ is associated with the m th candidate position. Also, let $\mathbf{d}(\omega_j, \theta_p)$ be an $M \times 1$ steering vector for a given $\theta_p \in \Theta$, $p = 1, \dots, P$, defined as $\mathbf{d}(\omega_j, \theta_p) = [d_1(\omega_j, \theta_p), \dots, d_M(\omega_j, \theta_p)]^T$ where Θ is the angle space, and the m th element is $d_m(\omega_j, \theta_p) = e^{i \frac{\omega_j}{c} (x_m \cos \theta_p + y_m \sin \theta_p)}$ [18] where (x_m, y_m) is the location of the m th sensor, and c is the speed of sound. The array BP for a given frequency ω_j and azimuth angle θ_p can be expressed as:

$$\mathcal{B}(\omega_j, \theta_p) = \mathbf{h}^H(\omega_j) \mathbf{d}(\omega_j, \theta_p). \quad (2)$$

Define the $M \times P'$ steering matrix associated with the mainlobe as $\mathbf{D}_{M, \Theta_m}(\omega_j) = [\mathbf{d}(\omega_j, \theta_1), \dots, \mathbf{d}(\omega_j, \theta_{P'})]$ and similarly define the matrix $\mathbf{D}_{M, \Theta_s}(\omega_j)$ associated with the sidelobes, and consider the following $P \times M$ dictionary matrices

$$\mathbf{D}_{\omega_j} = \begin{bmatrix} \mathbf{D}_{M, \Theta_m}^H(\omega_j) \\ \mathbf{D}_{M, \Theta_s}^H(\omega_j) \end{bmatrix}. \quad (3)$$

Each column of $\mathbf{D}_{\omega_j}$ is considered as an atom in the dictionary corresponding to one sensor at frequency ω_j . We then express the designed BP at a given frequency ω_j as a $P \times 1$ vector $\mathbf{b}(\omega_j) = \mathbf{D}_{\omega_j} \mathbf{h}(\omega_j)$. We want to express the desired BP $\mathbf{b}_d(\theta_{ds})$ with atoms from the dictionary $\mathbf{D}_{\omega_j}$ by finding vectors $\{\mathbf{h}(\omega_j)\}_{j=1}^J$ with the same support and that they are sparse vectors, that is, these vectors solve:

$$\min_{\{\mathbf{h}(\omega_j)\}_{j=1}^J} \sum_{j=1}^J \|\mathbf{h}(\omega_j)\|_0 \text{ s.t. } \mathbf{D}_{\omega_j} \mathbf{h}(\omega_j) = \mathbf{b}_d(\theta_{ds}) \quad (4)$$

where $\|\mathbf{h}(\omega_j)\|_0$ is the ℓ_0 -norm, i.e., the number of nonzero elements in the vector $\mathbf{h}(\omega_j)$. The design needs to ensure that all the filters $\{\mathbf{h}(\omega_j)\}_{j=1}^J$ have the same sparse pattern, i.e., the same sensors are chosen over the entire frequency band Ω . In order to apply such a constraint we may introduce the ℓ_{12} -norm of the $K \times L$ matrix $\mathbf{X}$ as $\|\mathbf{X}\|_{12} \triangleq \sum_{k=1}^K \|\mathbf{X}(\mathbf{k}, :)\|_2$ meaning that we apply ℓ_1 -norm to elements, where the k th element is the ℓ_2 -norm of the k th row of the matrix $\mathbf{X}$. In our case, we require that the following ℓ_{12} -norm is less than a given threshold η , that is, $\sum_{m=1}^M \sqrt{\|\bar{\mathbf{h}}(m)\|_2^2} \leq \eta$ where $\bar{\mathbf{h}}(m) = [h(m, \omega_1), \dots, h(m, \omega_J)]^T$ and $h(m, \omega_j)$ is the m th entry of $\mathbf{h}(\omega_j)$. Our goal is to design a sparse array with an FI BP that is approximately equal to $\mathbf{b}_d(\theta_{ds})$ under WNG and the distortionless response (DR) constraints. This can be expressed by the following sparse optimization problem:

$$\begin{aligned} \min_{\{\mathbf{h}(\omega_j)\}_{j=1}^J} \quad & \sum_{j=1}^J \|\mathbf{h}(\omega_j)\|_0 \text{ s.t.} \\ & \sum_{j=1}^J \|\mathbf{D}_{\omega_j} \mathbf{h}(\omega_j) - \mathbf{b}_d(\theta_{ds})\|^2 \leq \epsilon \\ & \text{WNG}(\mathbf{h}(\omega_j)) \geq \gamma(\omega_j)^{-1}, j = 1, \dots, J \\ & \text{DR}(\mathbf{h}(\omega_j)) = 1, j = 1, \dots, J \\ & \sum_{m=1}^M \sqrt{\|\bar{\mathbf{h}}(m)\|_2^2} \leq \eta, \end{aligned} \quad (5)$$

where $\gamma(\omega_j)$ is a parameter expressing the maximal allowed white noise output power for frequency ω_j . This is a coupled optimization problem over all the variables $\{\mathbf{h}(\omega_j)\}_{j=1}^J$, which is more difficult to solve, so we suggest a greedy solution.

3. A MODIFIED OLS METHOD

Applying the orthogonal least squares (OLS) algorithm directly to our problem is impractical because it treats each frequency independently. Considering the need for joint sparsity, we modify the OLS method to the correct problem.

First, we define an $M \times M$ diagonal mask matrix $\mathbf{W} = \text{diag}(w_1, w_2, \dots, w_M)$ as where $w_m \geq 0$ are weights associated with the m th sensor. The basic idea is to update each weight according to the iterative selection of sensors. At the l th iteration step, selecting a sensor out of the possible set of sensors in the optimization problem is based on two factors: (i) the reduction in the fit error obtained by selecting it, (ii) the cost of selecting it. To ensure that a sensor close to previously chosen sensors is not selected, we multiply its fit error by its current weight. Hence, if the sensor is close to previously chosen sensors, the product of the two terms will increase, making it not beneficial to choose it. This is a type of regularization that penalizes the fit error. Given that the sensor m^* is selected at the $(l-1)$ th iteration step, we then update the weights as:

$$\mathbf{W}^{(l)} = \mathbf{W}^{(l-1)} \text{diag}(\mathbf{m}_{l-1}), \quad (6)$$

where $\mathbf{m}_{l-1}$ contains attenuation factors in indices of the neighboring sensors to the current selected sensor m^* , and zero at index m^* . We initialize it as $\mathbf{W}^{(0)} = \mathbf{I}_M$.

Assume we are now at the $(l+1)$ th iteration step, and we want to find the next atom that most improves the full-OLS fit over the previously selected sensors. We achieve this by solving a least-squares problem using the previously selected sensors and one additional sensor from all the remaining atoms in the current iteration, and then taking the maximal projection. We are interested in obtaining a sparse array layout that is joint to all $\omega_j \in \Omega$. Let the $l \times 1$ index vector containing the selected sensor indices at the end of iteration l be denoted by $\mathbf{i}^{(l)}$. Define the $l \times M$ matrices

$$\mathbf{D}_{\omega_j}(\mathbf{i}^{(l)}, :) = \mathbf{T}_s(\mathbf{i}^{(l)}) \mathbf{D}_{\omega_j}, \quad (7)$$

that contain only the rows of $\mathbf{D}_{\omega_j}$ whose indices are specified by $\mathbf{i}^{(l)}$, where $\mathbf{T}_s(\mathbf{i}^{(l)})$ is an $l \times P$ selection matrix. We find the new sensor out of $M \setminus \mathbf{i}^{(l)}$ that minimizes the function:

$$m^* = \underset{m \in \mathbf{i}^{(l)}}{\text{argmin}} w_m^{(l)} \sum_{j=1}^J \min_{\mathbf{h}(\omega_j)} \|\mathbf{b}_d - \mathbf{D}_{\omega_j}(\mathbf{i}_m^{(l)}, :) \mathbf{h}(\omega_j)\|_2^2, \quad (8)$$

where $w_m^{(l)} = \mathbf{W}^{(l)}(m, m)$ is the (m, m) th entry of the matrix $\mathbf{W}^{(l)}$ which is the weight associated with the m th sensor at the l th iteration step, and $\mathbf{i}_m^{(l)} = [(\mathbf{i}^{(l)})^T, m]^T$, and $\mathbf{D}_{\omega_j}(\mathbf{i}_m^{(l)}, :)$ is the sub matrix of $\mathbf{D}_{\omega_j}$ containing the rows in the set $\mathbf{i}_m^{(l)}$. Let $\bar{\mathbf{D}}_{\omega_j} = \mathbf{D}_{\omega_j}(\mathbf{i}_m^{(l)}, :)$. The solution for $\mathbf{h}(\omega_j)$ is $\hat{\mathbf{h}}(\omega_j) = \bar{\mathbf{D}}_{\omega_j}^+ \mathbf{b}_d(\theta_{ds})$, where $\mathbf{X}^+$ is the pseudo-inverse of $\mathbf{X}$. Substituting $\hat{\mathbf{h}}(\omega_j)$ back into (8) yields that m^* minimizes

$$f(m) = w_m^{(l)} \cdot \sum_{j=1}^J \|\mathbf{P}_j^\perp(m) \mathbf{b}_d(\theta_{ds})\|^2 \quad (9)$$

where $\mathbf{P}_j^\perp(m) = \mathbf{I}_P - \mathbf{P}_j(m)$ is the complementary orthogonal projection matrix, and $\mathbf{P}_j(m) = \bar{\mathbf{D}}_{\omega_j} \bar{\mathbf{D}}_{\omega_j}^+$ is the projection matrix. Instead of computing the complementary orthogonal projection

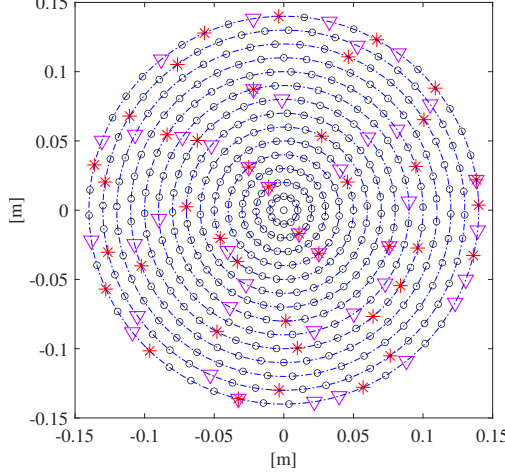


Fig. 1: Array layout obtained by the OLS design with 41 sensors (red stars), and the array layout obtained by the OMP design (purple triangles). The black circles are all the candidate sensors.

matrix for each candidate sensor location, and at each iteration, we modify the recursive implementation in [19, eq. (6)] to get that

$$f(m) = w_m^{(l)} \cdot \mathbf{b}_d^H(\theta_{ds}) \left[\sum_{j=1}^J \mathbf{P}_{j,i^{(l)}}^\perp \mathbf{D}_{\omega_j}(:, m) \mathbf{D}_{\omega_j}^H(:, m) \mathbf{P}_{j,i^{(l)}}^\perp \right] \mathbf{b}_d(\theta_{ds}) - \frac{\mathbf{P}_{j,i^{(l)}}^\perp \mathbf{D}_{\omega_j}(:, m) \mathbf{D}_{\omega_j}^H(:, m) \mathbf{P}_{j,i^{(l)}}^\perp}{\|\mathbf{P}_{j,i^{(l)}}^\perp \mathbf{D}_{\omega_j}(:, m)\|^2} \mathbf{b}_d(\theta_{ds}) \quad (10)$$

where $\mathbf{D}_{\omega_j}(:, m)$ is the m th column of $\mathbf{D}_{\omega_j}$. Then $\mathbf{P}_{j,i^{(l)}}^\perp$ is computed in an iterative way as [19, eq. (5)]

$$\mathbf{P}_{j,i^{(l)}}^\perp = \mathbf{P}_{j,i^{(l-1)}}^\perp + \frac{\mathbf{P}_{j,i^{(l-1)}}^\perp \mathbf{D}_{\omega_j}(:, m_l^*) \mathbf{D}_{\omega_j}^H(:, m_l^*) \mathbf{P}_{j,i^{(l-1)}}^\perp}{\|\mathbf{P}_{j,i^{(l-1)}}^\perp \mathbf{D}_{\omega_j}(:, m_l^*)\|^2} \quad (11)$$

where $\mathbf{P}_{j,i^{(l)}}^\perp$ is the complementary orthogonal projection matrix which is based on the matrix $\mathbf{D}_{\omega_j}(\mathbf{i}^{(l)}, :)$ in the previous step, and m_l^* is the selected sensor at the l th iteration step.

Since only the second term in (11) depends on m , substituting it into the minimization problem allows our objective function to be expressed as

$$m^* = \underset{m \in \{1, \dots, M\} \setminus \mathbf{i}^{(l)}}{\operatorname{argmin}} w_m^{(l)} \cdot \mathbf{b}_d^H(\theta_{ds}) \mathbf{G}_m \mathbf{b}_d(\theta_{ds}) \quad (12)$$

where

$$\mathbf{G}_m = \sum_{j=1}^J \frac{\mathbf{P}_{j,i^{(l)}}^\perp \mathbf{D}_{\omega_j}(:, m) \mathbf{D}_{\omega_j}^H(:, m) \mathbf{P}_{j,i^{(l)}}^\perp}{\|\mathbf{P}_{j,i^{(l)}}^\perp \mathbf{D}_{\omega_j}(:, m)\|^2}, \quad (13)$$

which is a $P \times P$ matrix. After selecting the new sensor, we update the masking matrix as detailed in (6). We also update the set of selected sensors as $\mathbf{i}^{(l+1)} = [(\mathbf{i}^{(l)})^T, m^*]^T$ which is a $(l+1) \times 1$ vector.

As a stopping criterion, we check whether the chosen sensors up to the $(l+1)$ th iteration comply with the constraints specified in the previous section. To do so, we may solve the following constrained

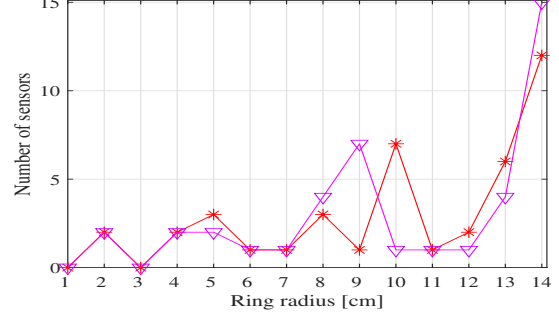


Fig. 2: Number of selected sensors in each ring obtained by the OLS design (red stars), and the OMP design (purple triangles).

optimization problem for each $\omega_j \in \Omega$ separately:

$$\begin{aligned} & \underset{\mathbf{h}_l(\omega_j)}{\operatorname{minimize}} \|\mathbf{h}_l(\omega_j)\|_2^2 \\ & \text{subject to} \\ & \mathbf{h}_l^H(\omega_j) \mathbf{T}_s(\mathbf{i}^{(l)}) \mathbf{d}(\omega_j, \theta_{ds}) = 1 \\ & \|(\mathbf{b}_d^m(\theta_{ds}))^T - \mathbf{h}_l^H(\omega_j) \mathbf{T}_s(\mathbf{i}^{(l)}) \mathbf{D}_{M, \Theta_m}(\omega_j)\|_2^2 \leq \epsilon_1(\omega_j) \\ & \|(\mathbf{b}_d^s(\theta_{ds}))^T - \mathbf{h}_l^H(\omega_j) \mathbf{T}_s(\mathbf{i}^{(l)}) \mathbf{D}_{M, \Theta_s}(\omega_j)\|_2^2 \leq \epsilon_2(\omega_j), \end{aligned} \quad (14)$$

If we get a valid solution to (14) and also $\|\mathbf{h}_l(\omega_j)\|_2^2 \leq \gamma(\omega_j)$, $\forall \omega_j \in \Omega$, then we stop, else, we repeat (8).

4. SIMULATION RESULTS

We consider a sparse design of a CCA, which is used in applications like video conference systems and voice service devices. Such geometry is popular in some practical applications requiring adaptive variable steering azimuthal direction of the mainlobe while preserving similar WNG and directivity pattern performance.

The CCA is composed of G rings, each characterized by its radius, r_g , and a number M_g which represents the maximum possible positions for locating sensors along the g th ring, where $g = 1, 2, \dots, G$. We assume $1 = M_1 \leq M_2 \leq \dots \leq M_G$. The total number of candidate positions is $M = \sum_{g=1}^G M_g$, and the center of the CCA coincides with the first ring, i.e., $r_1 = 0$ and $M_1 = 1$.

In the following, we focus on concentric circular differential microphone arrays (CCDMAs) originally designed with a linear geometry, which combines closely spaced sensors to respond to the spatial derivatives of the acoustic pressure field. These small-sized arrays yield nearly FI BPs and include the well-known superdirective beamformer as a particular case [20]. The general theoretical FI BP of an N th-order DMA is [21]

$$\mathcal{B}_d^{\theta_{ds}}(\theta) = \mathcal{B}_N(\theta_{ds} - \theta) = \sum_{n=0}^N a_{N,n} \cos^n(\theta_{ds} - \theta), \quad (15)$$

where $\{a_{N,n}\}_{n=0}^N$ are real coefficients, and the desired signal arrives from the direction θ_{ds} . The element spacing δ is required to be small enough so that $\delta \ll \lambda/2 = \frac{\pi c}{\omega}$, that is, $\delta \ll \frac{\pi c}{\omega_{\max}}$ for all $\omega \in \Omega$, where $\omega_{\max}$ is the angular frequency corresponding to the highest frequency in the bandwidth of interest Ω , and $c = 340$ m/sec. Therefore, $\delta = 1$ cm is satisfactory for our example.

$$\mathcal{B}_N^{\text{HC}}(\theta) = -0.14 - 0.57 \cos \theta + 0.57 \cos^2 \theta + 1.15 \cos^3 \theta. \quad (16)$$

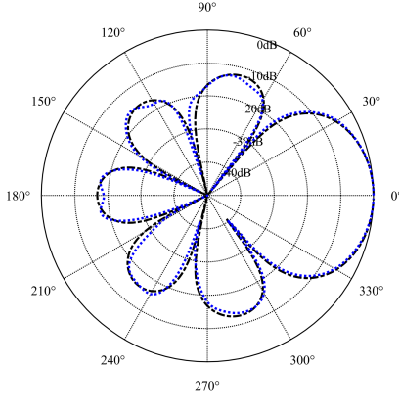


Fig. 3: Beampatterns of a 3rd-order hypercardioid for $f = 6280$ Hz, steered to $\theta_{ds} = 0^\circ$ (blue) obtained by the OLS design, and the theoretical beampattern of a 3rd-order hypercardioid (black dotted line) steered to $\theta_{ds} = 0^\circ$.

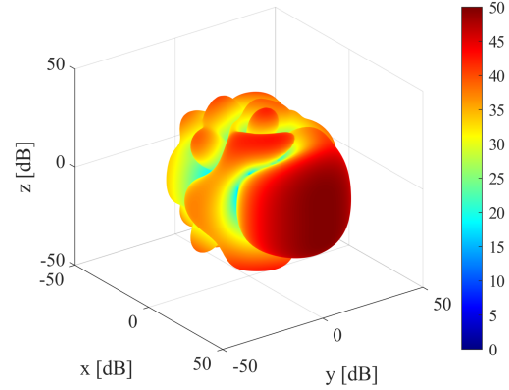
The angular axis is uniformly discretized with $\Delta\theta = 2^\circ$. For that case, we set $\theta_{P'} = 60^\circ$, i.e., the mainlobe region in the azimuthal axis is $-60^\circ \leq \theta \leq 60^\circ$. We set initial values for the tolerance parameters $\{\epsilon_1(\omega_j)\}_{j=1}^J$, $\{\epsilon_2(\omega_j)\}_{j=1}^J$, and $\{\gamma(\omega_j)\}_{j=1}^J$, by applying the parameters adjustment procedure introduced in [11].

Figure 1 presents the array layout obtained from the joint-sparse OLS algorithm proposed in Section 3 (marked by red stars). Also presented is the array layout obtained by the joint-sparse OPM algorithm we recently proposed in [13] (marked by purple triangles). Both algorithms yield a similar array layout with comparable performance. This similarity is also evident in Fig. 2, which shows the number of selected sensors per ring for both approaches. One can see that the sparse design offers a compromise between high and low frequencies and is rotationally invariant.

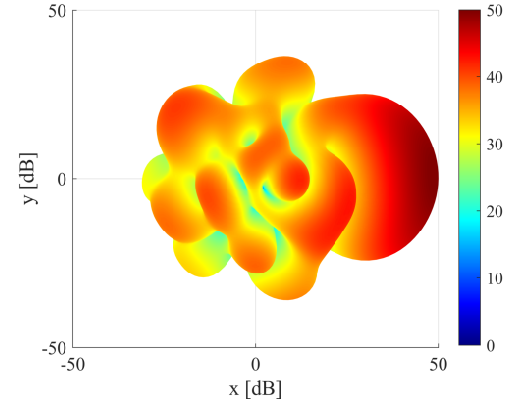
As both algorithms achieve similar performance, we may focus on the OLS results in the following. The FI BP, steered to $\theta_{ds} = 0^\circ$ (blue) by the proposed design, is shown in Fig. 3. The theoretical BP is also given (black dashed). Figure 4 shows the three-dimensional BP obtained by the proposed sparse design for two different angles of view. One can see the obtained mainlobe with low sidelobes. Figure 5 shows that the BP versus frequency received by the sparse array is practically FI, especially in the mainlobe region.

5. CONCLUSIONS

This paper addressed the challenges of synthesizing broadband FI beampatterns under strict hardware and computational constraints using sparse microphone arrays. We adapted the OLS algorithm into a joint-sparse optimization framework, creating an effective greedy method for sensor selection. This algorithm iteratively incorporates candidate sensors across the operational frequency subband to reduce fit error while adhering to key design constraints such as WNG and sidelobe suppression levels. To manage the high computational complexity of OLS evaluations, we derived a fast recursive implementation. Simulations with concentric circular arrays showed that our joint-sparse OLS framework provides robust, frequency-invariant, and rotationally-invariant directivity patterns, matching the performance of state-of-the-art OMP routines while offering greater design flexibility for advanced sparse arrays.



(a) side view



(b) top view

Fig. 4: 3D Beampatterns of a 3rd-order hypercardioid for $f = 6280$ Hz, steered to $\theta_{ds} = 0^\circ$ obtained by the OLS-based design, from side view (a) and top view (b).

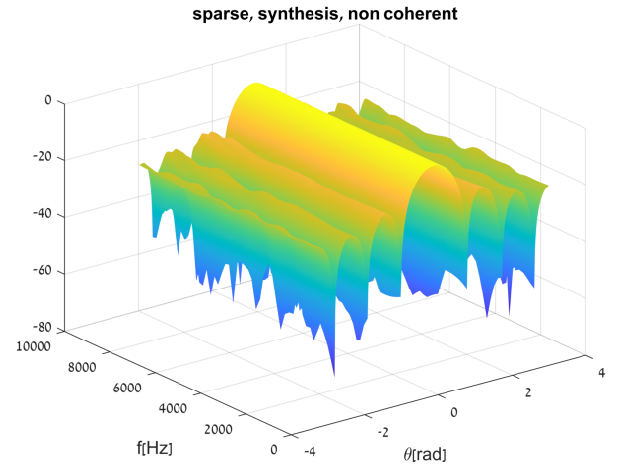


Fig. 5: Beampatterns versus frequency for the OLS-based design.

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